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Unit – 4 : Session – 12 : SLO – 1 | Hashing — Introduction

■ I. Fill in the Blanks

- Hashing is used to perform efficient SEARCHING (insertion/deletion/lookup).
- A HASH FUNCTION maps keys to indices.
- A good hash function distributes keys UNIFORMLY (evenly).
- When two keys hash to the same index, it's called a COLLISION.
- Collision RESOLUTION (handling) techniques handle situations when two keys hash to same index.
- OPEN ADDRESSING and CHAINING are two major collision resolution techniques.
- In open addressing, all elements are stored in the same TABLE (array).
- In chaining, each bucket contains a LINKED LIST.
- LOAD FACTOR is the ratio of number of elements to table size.
- A hash function should be EASY (fast) to compute.
- LINEAR PROBING handles collision by checking the next slot.
- Quadratic probing uses a QUADRATIC FUNCTION (i^2) to find next slot.
- DOUBLE HASHING uses a second hash function to calculate step size.
- Hash tables are widely used in DATABASES and COMPILERS (caching/dictionaries).
- The time complexity of hashing for search in average case is $O(1)$.